EE2001 Electrical Circuits- I:

Continuation of the Topic:
Sinosoidal Steady-State Analysis

02 January 2024

The Phasor

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$$V = V_m e^{j\phi}\ (polar\ form) = V_m cos\phi + jV_m sin\phi\ (rectangular\ form)$$

- Both polar and rectangular forms are useful in circuit applications of the phasor concept.
- The frequent use of the exponential function $e^{j\phi}$ has led to an abbreviation called as angle notation. We extensively use the folowing notation:

$$e^{j\phi} = 1\angle\phi$$

- Inverse phasor transform example:

$$V = 100\angle -26° = 100\text{cost}(\omega t - 26°)$$

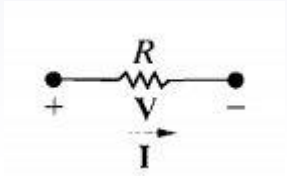
- The phasor transform is useful in circuit analysis, because it reduces the task of finding the maximum amplitude and phase angle of steady-state sinusoidal response to the algebra of complex numbers.
- The transient component vanishes as time elapses.
- Current can be expressed as $I_m e^{j\beta} = \frac{V_m e^{j\varphi}}{R + j\omega L}$

The Passive Circuit Elements in the Frequency Domain

The V-I relationship for a resistor:

- If the current in a resistor varies sinusoidally with time, that is, if

$$i(t) = I_m \cos(\omega t + \theta_i)$$



the voltage at the terminals of the resistor can be expressed as

$$v(t) = R[I_m \cos(\omega t + \theta_i)] = RI_m[\cos(\omega t + \theta_i)]$$

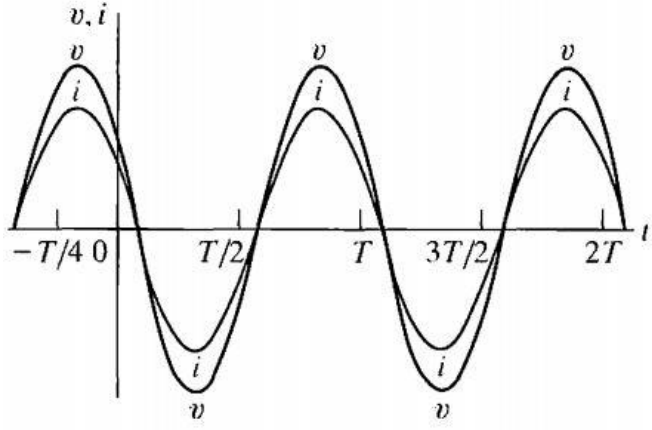
where I_m is the maximum amplitude of the current in amperes and θ_i is the phase angle of the current.

- The phasor transform of this voltage is

$$\boldsymbol{V} = RI_m e^{j\theta_i} = RI_m \angle\theta_i$$

- $I_m \angle\theta_i$ is the phasor representation of the sinusoidal current, so we can write the same equation as

$$\boldsymbol{V} = R\boldsymbol{I}$$

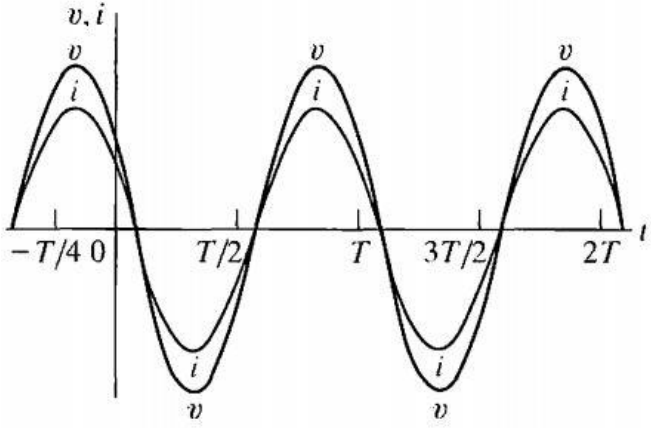


The Passive Circuit Elements in the Frequency Domain

The V-I relationship for a resistor:

$$V = RI$$

- This states that the phasor voltage at the terminals of a resistor is simply the resistance times the phasor current.
- There is no phase shift between the current voltage (they are in phase).
- In the given figure, the phase angle of both the voltage and the current waveforms is 60° They both reach corresponding values (such as positive maxima) on their respective curves at the same time.



The Passive Circuit Elements in the Frequency Domain

The V-I relationship for an inductor:

- Assuming a sinusoidal current and using $Ldi(t)/dt$

$$v(t) = L\frac{di(t)}{dt} = -\omega L I_m \, sin(\omega t + \theta_i)$$

- Rewriting using the cosine function

$$v(t) = -\omega L I_m \, cos(\omega t + \theta_i - 90°)$$

- The phasor representation of the voltage:

$$\boldsymbol{V} = -\omega L I_m e^{j(\theta_i - 90°)} = -\omega L I_m e^{j\theta_i} e^{j-90°} = j\omega L I_m e^{j\theta_i} = j\omega L \boldsymbol{I}$$

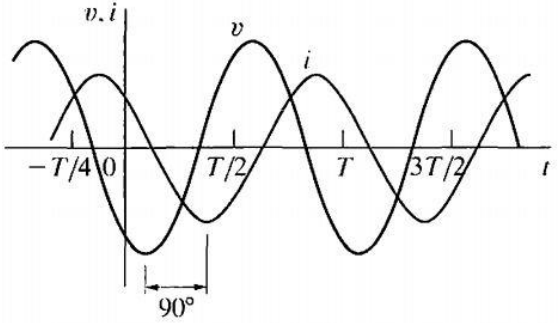
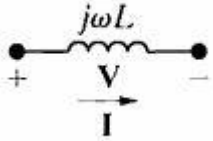


Figure 9.11 ▲ A plot showing the phase relationship between the current and voltage at the terminals of an inductor ($\theta_i = 60°$).

- Note that in deriving the equation we used the identity

$$e^{-j90°} = cos90° - jsin90° = -j$$



- Phasor voltage at the terminal of an inductor equals $j\omega L$ time the phasor current. This relationship applies as well for the mutual inductance in one coil due to current flowing in another mutually coupled coil. ($j\omega M$) times the phasor current in the other coil.

The Passive Circuit Elements in the Frequency Domain

The V-I relationship for an inductor:

$$\boldsymbol{V} = (\omega L \angle 90°) I_m \angle \theta_i = \omega L I_m \angle (\theta_i + 90)°$$

which indicates that the voltage and the current are out of phase by exactly 90°. In particular, the voltage leads the current by 90°, or equivalently, the current lags behind the voltage by 90°.

- We call this concept as voltage leading current, or current lagging voltage.
- For example, the voltage reaches its negative peak exactly 90° before the current reaches its negative peak. The same observation can be made with respect to the zero-going-positive crossing or the positive peak.
- Expressing this shift in seconds, 90° corresponds to one-fourth of a period; hence the voltage leads the current by $T/4$ or $\frac{1}{4f}$

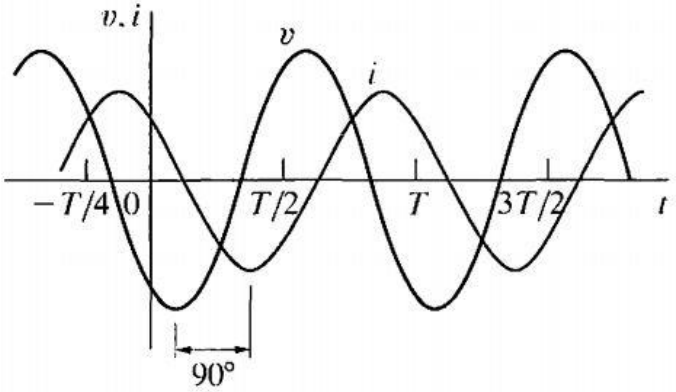


Figure 9.11 ▲ A plot showing the phase relationship between the current and voltage at the terminals of an inductor ($\theta_i = 60°$).

The Passive Circuit Elements in the Frequency Domain

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The V-I relationship for a capacitor:

- We note for a capacitor that $i(t) = Cdv(t)/dt$
- And assuming that: $v(t) = V_m \cos(\omega t + \theta_v)$
- Then

$$\boldsymbol{I} = j\omega C\boldsymbol{V}$$

Solving the equation for voltage we get:

$$\boldsymbol{V} = \frac{1}{j\omega C}\boldsymbol{I}$$

$$\boldsymbol{V} = \frac{1}{\omega C}\angle -90° I_m \angle \theta_i = \frac{I_m}{\omega C}\angle(\theta_i - 90)°$$

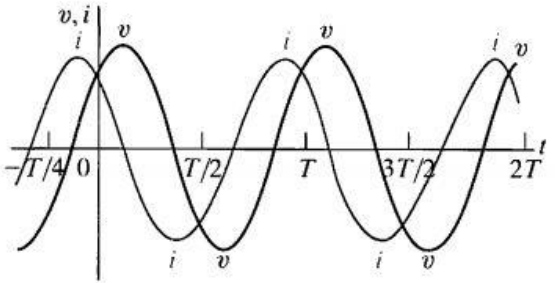
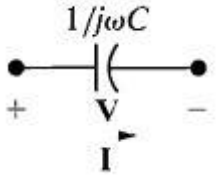


Figure 9.13 ▲ A plot showing the phase relationship between the current and voltage at the terminals of a capacitor ($\theta_i = 60°$).



- The voltage across the terminals of a capacitor lags behind the current by exactly 90°. We can alternatiely says that current leads the voltage by 90°. Figure shows this relationship.

4 The Passive Circuit Elements in the Frequency Domain

- The voltage and current relation of the passive circuit elements can be represented as:

$$\boldsymbol{V} = Z\boldsymbol{I}$$

- Z represents the impedance of the circuit element. It is the ratio of a circuit element's voltage phasor to its current phasor.
- The impedance of a resistor is R, the impedance of an inductor is $\mathrm{j}\omega L$, and the impedance of a capacitor is $-1/\mathrm{j}\omega C$. In all cases it is measured in ohms.
- Although impedance is a complex number, it is not a phasor. (phasor is a complex number that shows up as the coefficient of $e^{\mathrm{j}\omega t}$. Although all phasors are complex numbers, not all complex numbers are phasors.
- The imaginary part of the impedance is called reactance.

Kirchhoff's Laws in the Frequency Domain

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Kirchoff's Voltage Law (KVL) in the Frequency Domain

- Assuming that $v_1 - v_n$ represent voltages around a closed path in a circuit. Also assuming that the circuit is operating in a sinusoidal steady state. KVL requires that

$$v_1 + v_2 + \cdots + v_n = 0$$

which in the sinusoidal state becomes complex

$$V_{m_1}\cos(\omega t + \theta_1) + V_{m_2}\cos(\omega t + \theta_2) + \cdots + V_{m_n}\cos(\omega t + \theta_n) = 0$$

using Euler's identity,

$$Re\{V_{m_1}e^{j\theta_1}e^{j\omega t}\} + Re\{V_{m_2}e^{j\theta_2}e^{j\omega t}\} + \cdots + Re\{V_{m_n}e^{j\theta_n}e^{j\omega t}\} = 0$$

Factoring the term $e^{j\omega t}$ from each term yields

$$Re\{(V_{m_1}e^{j\theta_1} + V_{m_2}e^{j\theta_2} + \cdots + V_{m_n}e^{j\theta_n})e^{j\omega t}\} = 0$$

or

$$Re\{(\boldsymbol{V_1} + \boldsymbol{V_2} + \cdots + \boldsymbol{V_n})e^{j\omega t}\} = 0$$

Kirchhoff's Laws in the Frequency Domain

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$$Re\{(\boldsymbol{V_1} + \boldsymbol{V_2} + \cdots + \boldsymbol{V_n})e^{j\omega t}\} = 0$$

Because $e^{j\omega t} \neq 0$,

$$(\boldsymbol{V_1} + \boldsymbol{V_2} + \cdots + \boldsymbol{V_n}) = 0$$

which is the statement of KVL as it applies to phasor voltages.

Kirchoff's Current Law (KCL) in the Frequency Domain

- A similar derivation applies to a set of sinusoidal currents. Thus if

$$i_1 + i_2 + \cdots + i_n = 0$$

then

$$\boldsymbol{I_1} + \boldsymbol{I_2} + \cdots + \boldsymbol{I_n} = 0$$

Therefore you may use all the techniques developed for analyzing resistive circuits to find phasor currents and voltages. You need to learn no new analytic techniques, the basic circuit analysis and simplification tools covered in the previous topics can all be used to analyze circuits in the frequency domain.

5 Kirchhoff's Laws in the Frequency Domain

- Phasor circuit analysis consists of two fundamental tasks:
  - ➢ You must be able to construct the frequency-domain model of a circuit.
  - ➢ You must be able to manipulate complex numbers and/or quantities algebraically.

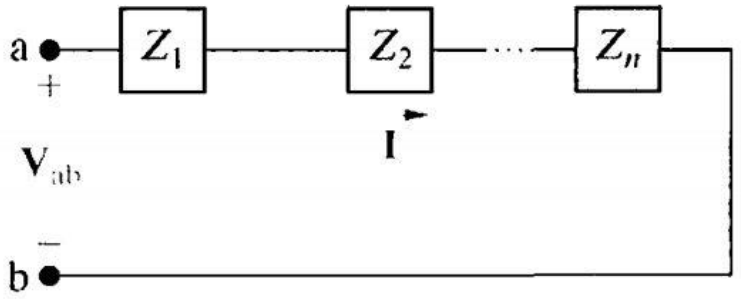
Series, Parallel and Delta-to-Wye Simplifications

- The rules for combining impedances in series or parallel and for making delta-to-wye transformations are the same as those for resistors. The only difference is that combining impedances involves algebraic manipulation of complex numbers.

Combining Impedances in Series and Parallel

- Impedances in series can be combined into a single impedance by simply adding the individual impedances.

$$\boldsymbol{V_{ab}} = Z_1\boldsymbol{I} + Z_2\boldsymbol{I} + \cdots + Z_n\boldsymbol{I}$$
$$= (Z_1 + Z_2 + \cdots + Z_n)\boldsymbol{I}$$



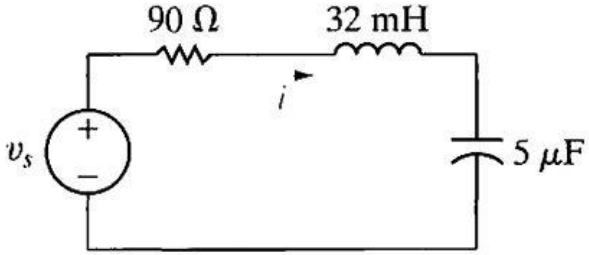
The equivalent impedance between terminals a and b,

$$Z_{ab} = \frac{\boldsymbol{V_{ab}}}{\boldsymbol{I}} = Z_1 + Z_2 + \cdots + Z_n$$

Series, Parallel and Delta-to-Wye Simplifications

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- Example-1: A 90 Ω resistor, a 32 mH inductor, and a 5 μF capacitor are connected in series across the terminals of a sinusoidal voltage source in the given circuit. The steady-state expression for the source voltage v_s is $750\cos(5000t + 30°)\ V$.



a) Construct the frequency-domain equivalent circuit.

b) Calculate the steady-state current i by the phasor method.

Solution: a) From the expression for v_s, we have $\omega = 5000\ rad/s$. Therefore, the impedance of the 32 mH inductor is

$$Z_L = j\omega L = j(5000)(32 \times 10^{-3}) = j160\ \Omega$$

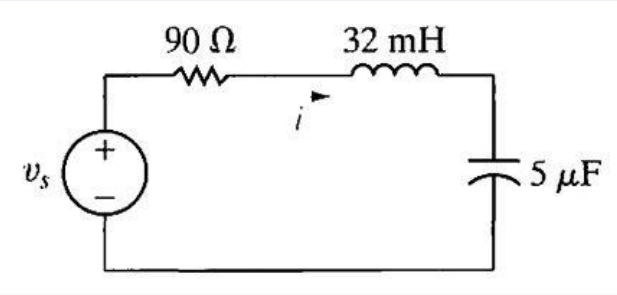
and the impedance of the capacitor is

$$Z_C = j\frac{-1}{\omega C} = -j\frac{10^6}{(5000)(5)} = -j40\ \Omega$$

Series, Parallel and Delta-to-Wye Simplifications

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- Example-1: A 90 Ω resistor, a 32 mH inductor, and a 5 μF capacitor are connected in series across the terminals of a sinusoidal voltage source in the given circuit. The steady-state expression for the source voltage v_s is $750\cos(5000t + 30°)\ V$.



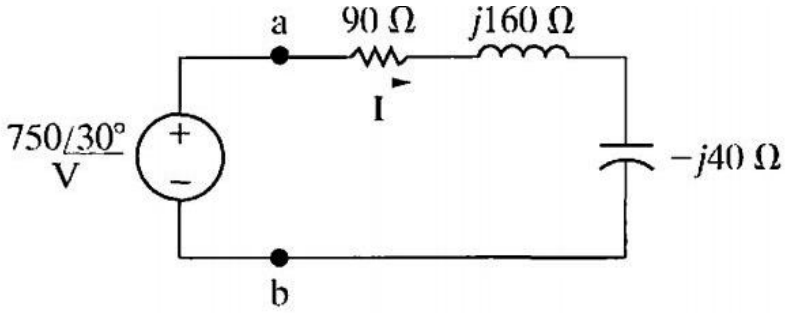
a) Construct the frequency-domain equivalent circuit.

b) Calculate the steady-state current i by the phasor method.

Continuation of the solution-a: The phasor transform of v_s is

$$\boldsymbol{V_s} = 750\angle 30°$$

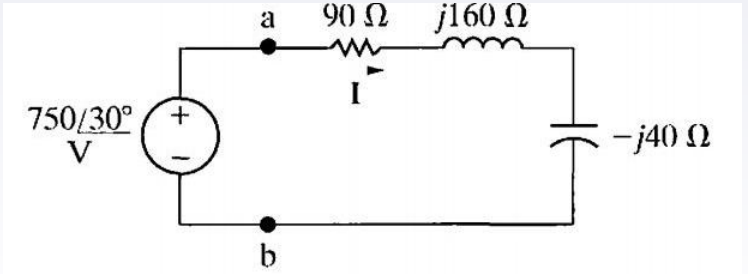
The frequency-domain equivalent circuit of the given circuit:



Series, Parallel and Delta-to-Wye Simplifications

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- Example-1: A 90 Ω resistor, a 32 mH inductor, and a 5 μF capacitor are connected in series across the terminals of a sinusoidal voltage source in the given circuit. The steady-state expression for the source voltage v_s is $750\cos(5000t + 30°)\ V$.



b) Calculate the steady-state current i by the phasor method.

Solution-b: We compute the phasor current simply by dividing the voltage of the voltage source by the equivalent impedance between the terminals a and b.

$$Z_{ab} = 90 + j160 - j40 = 90 + \text{j}120 = 150\angle 53.13°\ \Omega$$

Thus

$$\boldsymbol{I} = \frac{750\angle 30°}{150\angle 53.13°} = 5\angle -23.13°\ A$$

We may now write the steady-state expression for i directly:

$$i(t) = 5cos(5000t - 23.13°)\ A$$

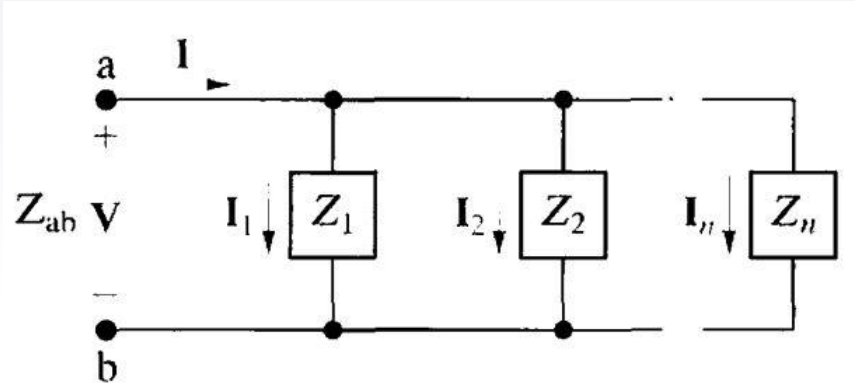
Series, Parallel and Delta-to-Wye Simplifications

Combining Impedances in Series and Parallel

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- Impedances connected in parallel may be reduced to a single equivalent impedance by the reciprocal relationship:

$$\frac{1}{Z_{ab}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_n}$$



- When impedances are in parallel, they have the same voltage across their terminals.
- By simply combining KCL with the phasor-domain version of Ohm's law,

$$I = I_1 + I_2 + \cdots + I_n \quad \text{or} \quad \frac{V}{Z_{ab}} = \frac{V}{Z_1} + \frac{V}{Z_2} + \cdots + \frac{V}{Z_n}$$

- Cancelling the common voltage term out of the equation, an equivalent impedance for the case of just two impedances in parallel

$$Z_{ab} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

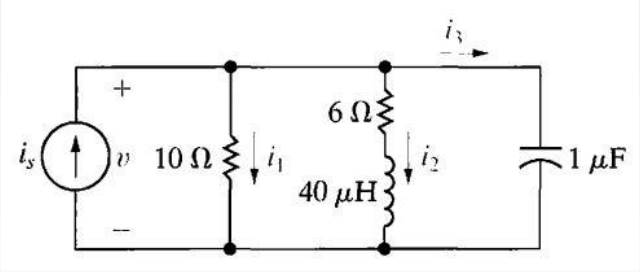
Series, Parallel and Delta-to-Wye Simplifications

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- Example-2: The sinusoidal current source in the given circuit produces the current $i_s = 8\cos 200000t$ A.

  a) Construct the frequency-domain equivalent circuit.

  b) Calculate the steady-state v, i_1, i_2 and i_3.

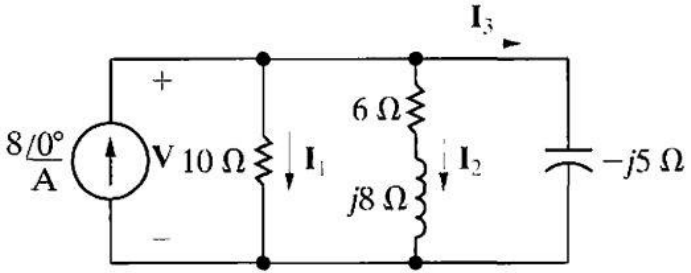


Solution: a) The phasor transform of the current source is

$$\boldsymbol{I_s = 8\angle 0°}$$

The resistors transform directly to the frequency domain as 10 and 6 Ω, the 40 μH inductor has an impedance of $j8$ Ω at the given freqeuncy of 200000 rad/s and 1 μF capacitor has an impedance of $-j5$ Ω at this freqeuncy.

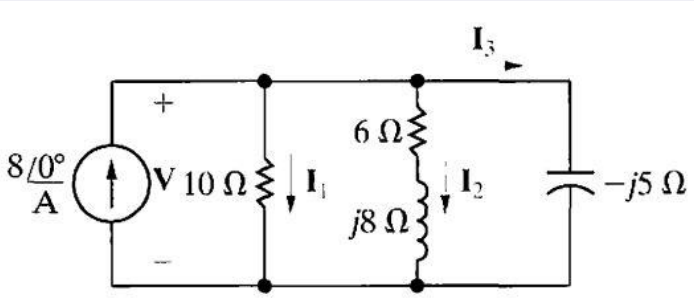
The frequency-domain equivalent circut and symbols representing the phasor transform of unknowns:



Series, Parallel and Delta-to-Wye Simplifications

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- Example-2: The sinusoidal current source in the given circuit produces the current $i_s = 8\cos 200000t$ A.



b) Calculate the steady-state current v, i_1, i_2 and i_3.

Solution: b) We can easily obtain the voltage across the current source once we know the equivalent impedance of the three parallel branches. Once we know $\boldsymbol{V}$, we can calculate the three phasor currents $\boldsymbol{I_1}$, $\boldsymbol{I_2}$ and $\boldsymbol{I_3}$. To find the equivalent impedance of the three branches, we first find the equivalent admittance simply by adding the admittances of each branch.

The admittance of the first branch is

$$Y_1 = \frac{1}{10} = 0.1\ S$$

The admittance of the second branch is

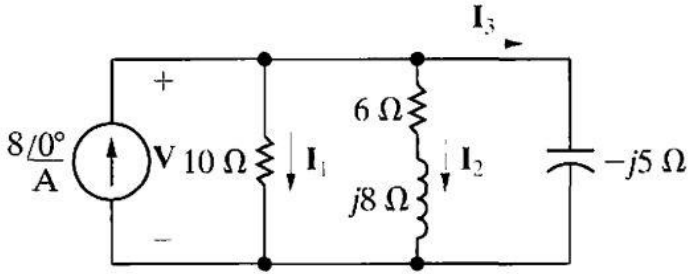
$$Y_2 = \frac{1}{6 + j8} = \frac{6 - j8}{100} = 0.06 - j0.08\ S$$

Series, Parallel and Delta-to-Wye Simplifications

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- Example-2: The sinusoidal current source in the given circuit produces the current $i_s = 8\cos 200000t$ A.

  b) Calculate the steady-state current v, i_1, i_2 and i_3.



Solution: b) The admittance of the third branch is

$$Y_3 = \frac{1}{-j5} = j0.2\ S$$

The admittance of the three branches is

$$Y = Y_1 + Y_2 + Y_3 = 0.16 + j0.12 = 0.2\angle 36.87\,°\,\mathrm{S}$$

The impedance at the current source is

$$Z = \frac{1}{Y} = 5\angle -36.87\,°\,\Omega$$

The Voltage **V** is

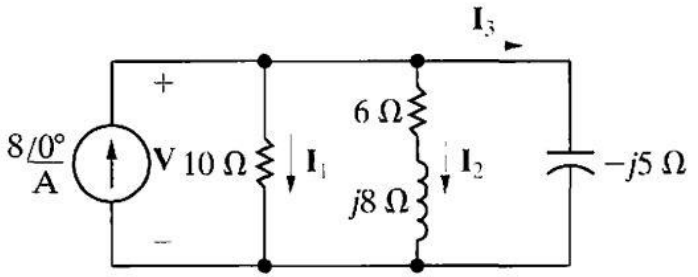
$$\boldsymbol{V} = Z\boldsymbol{I} = 40\angle -36.87\,°\,\mathrm{V}$$

Series, Parallel and Delta-to-Wye Simplifications

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- Example-2: The sinusoidal current source in the given circuit produces the current $i_s = 8\cos 200000t$ A.

  b) Calculate the steady-state current v, i_1, i_2 and i_3.



Conituation of solution b: $\boldsymbol{V = ZI} = 40\angle -36.87\,°\,\text{V}$

Hence,

$$\boldsymbol{I_1} = \frac{40\angle -36.87\,°}{10} = 4\angle -36.87\,° = 3.2 - \text{j}2.4\ \text{A}$$

$$\boldsymbol{I_2} = \frac{40\angle -36.87\,°}{6 + j8} = 4\angle -90\,° = -\text{j}4\ \text{A}$$

$$\boldsymbol{I_3} = \frac{40\angle -36.87\,°}{5\angle -90\,°} = 8\angle 53.13\,° = 4.8 - \text{j}6.4\ \text{A}$$

We check the computations at this point by verifying that

$$\boldsymbol{I_1 + I_2 + I_3 = I}$$

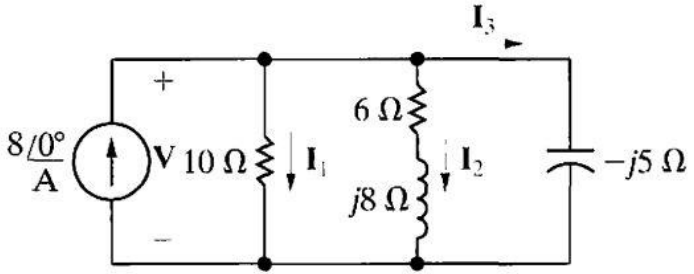
$$3.2 - \text{j}2.4 - \text{j}4 + 4.8 - \text{j}6.4 = 8 + \text{j}0$$

Series, Parallel and Delta-to-Wye Simplifications

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- Example-2: The sinusoidal current source in the given circuit produces the current $i_s = 8\cos 200000t$ A.

  b) Calculate the steady-state current v, i_1, i_2 and i_3.



Conituation of solution b:

The corresponding steady-state time-domain expressions are

$$v(t) = 40\cos(200000t - 36.87\,°)\,V$$
$$i_1(t) = 4\cos(200000t - 36.87\,°)\;A$$
$$i_2(t) = 4\cos(200000t - 90\,°)\;A$$
$$i_3(t) = 8\cos(200000t + 53.13\,°)\;A$$

Series, Parallel and Delta-to-Wye Simplifications

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- Admittance is the reciprocal of impedance and denoted Y.

$$Y = \frac{1}{Z} = G + jB\ (siemens) \qquad G = \frac{1}{R}$$

- Admittance is a complex number too, whose real part, G, is called conductance and whose imaginary part, B, is called susceptance. They are all measured in siemens (S).
- The equivalent admittance of parallel connected passive circuit elements can be calculated as

$$Y_{ab} = Y_1 + Y_2 + \cdots + Y_n$$

- The admittance of each of the ideal passie circuit elements is worth noting and summarized below:

Circuit Element	**Admittance (Y)**	**Susceptance**
Resistor	G (conductance)	None
Inductor	$j(-1/j\omega L)$	$-1/j\omega L$
Capacitor	$j\omega C$	ωC

End of the Lesson (02.01.24)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist